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Case Study: NLP for Quality Control Reports in Production

1. Project Overview

Course: MDM-09 — Case Study: Intelligent Systems in Production
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This project explores the integration of Natural Language Processing (NLP) and Data visualisation techniques to automate quality inspection processes within production lines. It focuses on automating the identification of product defects using image classification models and analyzing text-based defect reports to uncover recurring issues and improve manufacturing efficiency.

2. Research Context and Motivation

Traditional quality inspection systems in manufacturing often rely on manual observation and documentation, which are prone to human error, inefficiency, and high operational costs. The team identified a research gap: few existing studies successfully combine NLP and Computer Vision into a single automated pipeline, particularly for small and medium-sized enterprises (SMEs) within the food and beverage industry.

Additionally, limited availability of open datasets for defect detection in consumer products hinders rapid prototyping and scalability. The project aims to address this challenge by building an easy-to-deploy AI prototype that leverages existing data, simple interfaces, and low-cost sensors or cameras.

3. Literature and Conceptual Framework

The literature review highlights three major thematic areas that shape the foundation of this project:

Data Visualisation and Defect Detection:
Recent advancements in computer vision, particularly through Convolutional Neural Networks (CNNs), have revolutionized automated inspection systems. CNNs are capable of learning hierarchical image features that allow them to detect subtle manufacturing defects such as cracks, scratches, or discolorations. Transfer learning, where pretrained models like ResNet or VGG are adapted to domain-specific datasets, has proven especially effective in scenarios with limited labeled data. These studies establish the feasibility of real-time image-based quality assurance.

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NLP for Quality Report Analysis:
Natural Language Processing (NLP) techniques have been increasingly adopted in industrial analytics to interpret unstructured text data such as maintenance logs, defect reports, and operator comments. Techniques including tokenization, stopword removal, lemmatization, and keyword extraction help identify recurring issues and root causes. Text mining enables the conversion of qualitative data into actionable insights, enhancing decision-making in manufacturing contexts.

Integrated AI Pipelines:
Although computer vision and NLP have been individually explored for quality control, research combining both modalities remains limited, often restricted to large-scale industries. Early works on multimodal AI pipelines suggest that integrating visual and textual data can provide a more holistic view of production quality, leading to smarter, adaptive systems.

Project Contribution:
This project advances existing research by designing a unified, multimodal AI system that concurrently processes image and text data streams. By integrating CNN-based visual inspection with NLP-driven report analysis, the system aims to improve real-time decision-making and root-cause identification in manufacturing environments.

4. Methodology

4.1 Data Processing and Model Training

Dataset Upload & Preparation:
Custom scripts were developed to automatically organize image datasets into labeled categories (e.g., defective vs. non-defective). Data augmentation techniques—such as rotation, flipping, and brightness adjustment—were applied to expand dataset diversity and mitigate overfitting.

Model Architecture:
A baseline CNN model was implemented as the core of the defect detection module. The architecture consists of convolutional layers for feature extraction, max-pooling layers for dimensionality reduction, and fully connected layers for classification. An architecture summary and training visualization were generated using Keras/TensorFlow tools.

Training Process:
The model was trained using standard backpropagation with the Adam optimizer and categorical cross-entropy loss. Training produced interpretable metrics such as accuracy and loss curves, which stabilized around ~50% baseline accuracy, indicating room for further optimization.

Evaluation:
A confusion matrix was generated to evaluate performance across defect categories. Misclassifications were analyzed to identify biases and opportunities for model improvement.

4.2 NLP Module Design

Input:
The NLP module processes textual inputs including defect reports, maintenance logs, and operator notes, which often contain valuable contextual information on recurring issues.

Preprocessing:
Data cleaning involved tokenization, stopword removal, and lemmatization using NLTK and spaCy libraries. These steps standardize the text for downstream analysis and enhance keyword accuracy.

Analysis:
Statistical and linguistic analyses were performed to extract frequent keywords and generate word clouds, revealing dominant themes such as recurring machine faults or material inconsistencies.

Goal:
The NLP pipeline aims to enable root-cause identification by detecting trends and patterns in defect occurrences, thereby supporting preventive measures and process optimization.

5. Prototype Development

An early CAD design and system prototype were proposed to simulate the integrated AI quality control process.

Components:
The system includes a webcam for visual input, a conveyor belt to simulate continuous production flow, an AI model for defect classification, and a dashboard for visual analytics and alerts.

Simulation Data:
To demonstrate feasibility, open-source factory video datasets were used as proof-of-concept data streams.

Functionality:
The prototype successfully demonstrated real-time defect detection, generating instant alerts upon identifying defects. The design emphasizes low-cost scalability, making it accessible for small and medium manufacturing enterprises.

6. Project Management

Team Roles:

- Boshi Zhao: Led literature research and dataset preparation, ensuring data consistency and relevance.
- Teja Tadisetty: Focused on programming and developing the baseline CNN model.
- Anandu Hari: Managed system integration, analysis, and performance evaluation.

Timeline:
A Gantt chart was used to track project progress from October 16 onwards. Completed milestones include dataset preparation and baseline model training. The next phase prioritizes NLP integration and end-to-end system testing.

7. Issues Faced and Resolutions

Data Availability:
Limited access to proprietary manufacturing data was addressed by using open datasets and generating synthetic defect images to expand the dataset.

Model Overfitting:
Early experiments revealed overfitting due to limited data variety. Data augmentation and regularization techniques (e.g., dropout) were applied to improve generalization.

Integration Complexity:
Combining vision and NLP pipelines initially introduced synchronization challenges. This was resolved through a modular design, allowing independent testing and staged integration of components.

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8. Business and Industrial Impact

Factory Owner Benefits:

- Reduced Manual Inspection Costs: Automating defect detection minimizes the need for human inspectors.
- Minimized Rework and Scrap: Early defect detection prevents faulty products from advancing down the production line.
- Improved Production Speed and Quality: Continuous, real-time monitoring enhances throughput and consistency.
- Clear ROI: Reduced downtime and labor costs lead to tangible efficiency gains and measurable returns on investment.

Overall, the system represents a scalable, cost-effective solution that can bring Industry 4.0 capabilities to smaller manufacturers.

9. Conclusion and Future Work

This project establishes a foundation for an intelligent, multimodal quality control system capable of analyzing both visual and textual data streams. The integrated approach enhances decision-making, reduces waste, and increases production reliability.

Future Work:

- Dataset Expansion: Incorporate diverse defect types and more complex manufacturing scenarios.
- Deployment in Real Environments: Test the system in actual factory settings for validation.
- Integration with Predictive Maintenance: Link quality data with equipment health metrics for proactive maintenance scheduling.

By extending this framework, the project envisions a fully autonomous, adaptive quality assurance system for smart manufacturing environments.

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